Cat plot

- Figure-level interface for drawing categorical plots onto a FacetGrid.
- This function provides access to several axes-level functions that show the relationship between a numerical and one or more categorical variables using one of several visual representations.
- The kind parameter selects the underlying axes-level function to use.

Categorical scatterplots:

- stripplot() (with kind="strip"; the default)
- swarmplot() (with kind="swarm")

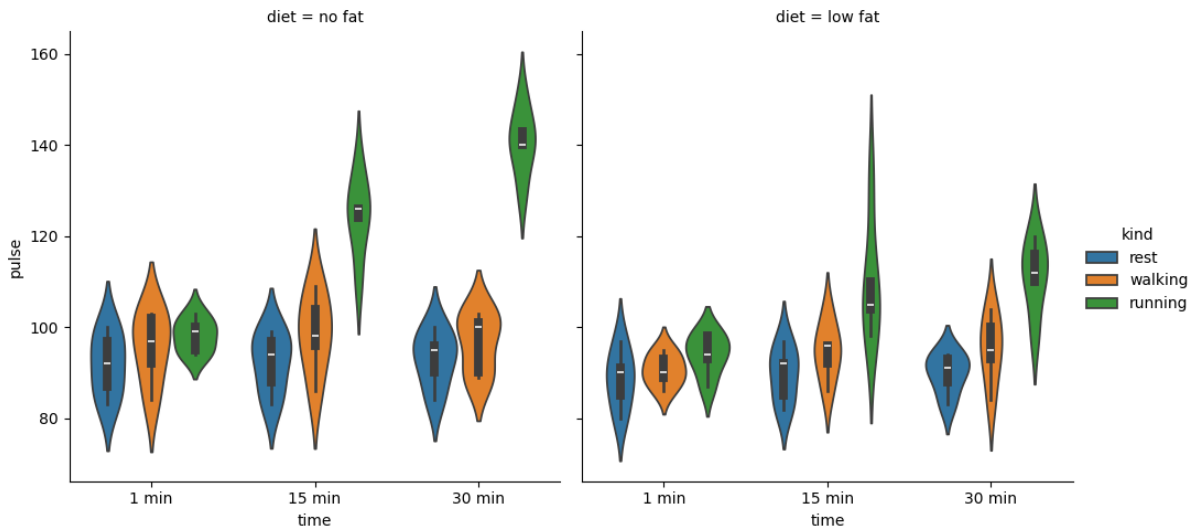
Categorical distribution plots:

- boxplot() (with kind="box")
- violinplot() (with kind="violin")
- boxenplot() (with kind="boxen")

Categorical estimate plots:

- pointplot() (with kind="point")
- barplot() (with kind="bar")
- countplot() (with kind="count")

Summary of cat plot:



Axes & Variables

- X-axis (time): 1 min, 15 min, 30 min.
- Y-axis (pulse): Heart rate (pulse).
- Hue (color / kind): Activity type (rest = blue, walking = orange, running = green).
- Columns (diet): Two panels show diets

    1. no fat (left)
    2. low fat (right).

Interpretation by conditions

1. No Fat Diet (Left Panel)

- Rest (blue): Pulse is lowest (around 85–100). Distribution is narrow and stable heart rate.
- Walking (orange): Pulse slightly higher than rest (90–105). Still stable.

- Running (green): Pulse much higher (120–150), increasing over time. At 30 min, highest values (~140+).

Observation: Running under no-fat diet causes a clear rise in pulse over time.

2. Low Fat Diet (Right Panel)

- Rest (blue): Pulse is lowest, around 80–95.
- Walking (orange): Slightly higher, similar to no-fat diet.
- Running (green): Pulse increases but shows more variation (wider violins). At 15 min, pulses are spread (100–150), and at 30 min, they reach ~120–135.

Observation: Under low-fat diet, running also increases pulse, but distribution is more spread out and thus individuals respond differently.

Overall Insights

1. Activity matters more than diet → Pulse increases from rest → walking → running in both diets.
2. Time effect: Longer activity = higher pulse, especially for running.
3. Diet effect:
    - No-fat diet → running pulse is consistently high and less variable.
    - Low-fat diet → running pulse shows more variability, especially at 15 min.
4. Resting & walking pulses are stable and similar across both diets.